ADAS systems must be engineered to handle edge cases with the same level of confidence and reliability as routine driving scenarios.

The solutions: A system-level rethink

The path forward is not about making incremental adjustments to existing systems. It requires rethinking the underlying foundations of ADAS design.

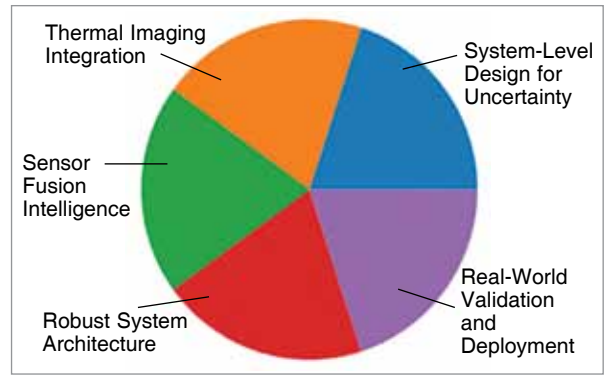
Moving beyond legacy thinking: Designing for uncertainty, not just performance. Improving ADAS reliability is not about incrementally refining individual components; it requires a shift towards system-level design that anticipates uncertainty. Traditional approaches optimise for average conditions. Real-world driving, however, is defined by edge cases, low visibility, ambiguous objects, and conflicting sensor inputs. Next-generation ADAS systems must therefore:

- Assume imperfect perception as the norm
- Prioritise robustness over peak-case accuracy
- Treat sensor diversity as a core design principle rather than an enhancement

This is where expanding beyond conventional sensing modalities becomes critical. This is the perfect point where thermal imaging fits in the ADAS stack.

Thermal imaging introduces a fundamentally different sensing modality by detecting heat signatures rather than reflected light. This makes it uniquely effective in situations where traditional sensors struggle. Key functional advantages:

- Low-light and night performance: Detects pedestrians, cyclists, and animals beyond headlight range, where cameras degrade significantly
- Glare immunity: Unaffected by sunlight or oncoming headlights—common triggers for perception errors
- Improved detection in partial occlusion: Heat signatures can remain distinguishable even when objects are visually obscured



ADAS solutions overview

- Reduced visual ambiguity: Minimises false positives caused by shadows, reflections, or low-contrast objects

Phantom braking is rarely the result of a random malfunction. More commonly, it arises from uncertainty or conflict within a vehicle's perception stack. When sensors deliver ambiguous or contradictory inputs, the system often defaults to caution, triggering unnecessary braking events. Thermal imaging can play a strategic role in mitigating this issue. By introducing an orthogonal layer of perception, thermal sensing enables cross-validation alongside conventional vision systems and provides an additional means of interpreting the environment. In complex or ambiguous situations, thermal sensing can help distinguish genuine obstacles from false positives. The result is a more balanced perception framework that reduces the likelihood of misclassification and, consequently, minimises phantom braking events.

Sensor fusion as a decision engine, not just data aggregation. Sensor fusion is often framed as merely combining inputs. In reality, its effectiveness is defined by how intelligently a system resolves conflict between those inputs. A modern fusion stack must go beyond aggregation, assigning confidence scores to each sensor stream, arbitrating contextually when inputs diverge, and prioritising complementary modalities over mere redundancy. It is not about having more data; it is about knowing which data to trust, and when.

This is where thermal imaging further strengthens the perception stack. Cameras provide semantic understanding. Radar measures distance and velocity. Lidar maps geometry and structure. Thermal sensing contributes a different layer of information by detecting heat signatures. This distinction is important because it introduces a layer of awareness that remains resilient when visibility deteriorates or conventional cues become unreliable.

In practice, thermal sensing delivers particular value during periods of low visual confidence, in validating edge cases, and in reducing false positives, especially within systems such as automatic emergency braking (AEB), where misinterpretation can trigger unnecessary intervention. The result is not only enhanced perception but also more consistent and dependable decision-making under uncertainty.

Thermal imaging is not, however, a plug-and-play upgrade. Its integration introduces additional design considerations. Cost implications must be balanced against measurable safety benefits. System complexity increases, requiring careful calibration and alignment with existing sensors. Additional data streams increase computational and synchronisation demands. Sensor placement also becomes critical, as field of view and environmental exposure directly affect performance.

The objective is not replacement but reinforcement. Thermal imaging is most effective when deployed strategically to strengthen vulnerable areas of the perception stack, improving both system capability and confidence.

System architecture: Enabling real-time, reliable decisions. Even the most advanced sensors remain